

ADITYA DWIVEDI

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BRIEF SUMMARY

- Final-year B.E. Computer Engineering student with hands-on experience in Linux, shell scripting, C++, Python, and system-level development. Skilled in data handling, CLI tooling, and research-driven engineering with strong collaboration and project ownership.

KEY EXPERTISE

- Linux, Shell Scripting, C++, Python, PHP, System Programming, CLI Tooling, Git, Data Handling, Research Workflows.

EDUCATION

Pune University

B.E. Computer Engineering

2026

Pune (M.H), India

Holy Cross S.S. School, Kapa

CBSE Senior Secondary Education

2019 – 2021

Raipur (C.G), India

RESEARCH / WHITE PAPERS

AI-Guided Memory Allocation for Edge AI Inference on Heterogeneous Systems

(Ongoing)

- Conducting research on AI-driven predictive memory allocation techniques to optimize inference performance on resource-constrained heterogeneous systems.
- Reviewing literature on dynamic memory management, edge AI inference, and heterogeneous computing architectures.
- Designing and planning experimental setups for software-based optimization without dedicated GPU resources.
- Preparing a full paper for submission to an IEEE conference, with expected completion in June 2026.

PROJECTS

Telos Behavioral Progress Tracker CLI & Mobile App)

July 2025

Key Skills - Python, Game Theory, CLI Systems, Flutter, Data Modeling

- Designed a behavioral tracking system predicting success probability using a hybrid Game Theory model.
- Built a CLI prototype in Python with monthly updates, trend detection, and custom advice engine.
- Led Flutter mobile app development with persistence, onboarding, and feedback generation.
- 🔗 github.com/logicnotfound237/Telos

Key Skills - C++, JSON, Shell, CLI UX, System Design

- Created a minimalist C++ based task manager for terminal productivity.
- Implemented JSON persistence and intuitive command-line interactions.
- Focused on lightweight performance and clean code structure for maintainability.
-  github.com/logicnotfound237/dodone

PAYLOAD ENGINEER STES Rocketry

Feb 2023 – July 2024

Key Skills - Aerospace Systems, Payload Design, Data Analysis, Experimental Engineering

- Designing and developing payloads for experimental rocket launches while ensuring compliance with mission objectives and technical specifications.
- Conducted extensive experimentation and data analysis to characterize the behaviour of levitated particles, investigating pressure distribution..
- Presented research findings at seminar, disseminating knowledge and promoting the potential of acoustic levitation technology in diverse fields.

SEMINARS / TRAININGS / WORKSHOPS

Brain-machine interface (BMI) integration with AI

Key Skills - Neuroscience, Computer Science [Interdisciplinary..Collaboration]

- The seminar on Brain-Machine Interface (BMI) explored the convergence of neuroscience, engineering, and AI. Attendees gained insights into the interdisciplinary skills required for BMI research, including neuroscience, engineering, and computer science. A key focus was on comparing different AI algorithms used in BMIs, providing attendees with valuable insights into their strengths and applications, fostering informed decision-making and innovation in BMI technology.

INTERNSHIP

Backend Developer Intern

Dec 2023 – Jan 2024

HIRA FERRO ALLOYS

Raipur

- Worked on system-level tooling and automation using Python and shell scripting to improve developer workflows.
- Assisted in building internal CLI utilities for data processing and reporting.
- Collaborated with senior engineers on debugging, optimization, and feature implementation.

CO-CURRICULAR ACTIVITIES

- Awarded Best NCC Cadet.
- Ex-Payload Engineer at STES Rocketry contributing to launch systems and experimental payload platforms.
- 2nd Place in International Latin America Space Challenge.